CAUSALITY ASSESSMENT

Pharmacovigilance | B.Pharm Study Notes | Simple English

1. Meaning of Causality Assessment

Causality assessment is the systematic process of evaluating the relationship between a suspected medicine and an adverse drug reaction (ADR). It helps determine how likely it is that the medicine caused the reaction.

In simple words: when a patient develops an unwanted effect after taking a medicine, causality assessment checks whether the medicine may have caused that effect or whether another factor is responsible.

2. Objectives

- To assess the likelihood that a suspected drug caused an ADR.
- To support accurate ADR documentation and reporting.
- To identify possible medicine-related safety problems.
- To support safer use of medicines and better patient care.
- To help pharmacovigilance teams evaluate safety signals.

3. Important Terms

Adverse Drug Reaction (ADR)

A harmful and unintended response to a medicine when it is used at normally prescribed doses for prevention, diagnosis, treatment, or modification of a physiological function.

Adverse Event (AE)

Any unwanted medical occurrence after the use of a medicine. It does not necessarily have a proven causal relationship with the medicine.

Suspected drug

The medicine that is considered a possible cause of the reported reaction.

Dechallenge

The suspected medicine is stopped or its dose is reduced, and the reaction is observed. Improvement after stopping the medicine may support a causal relationship.

Rechallenge

The medicine is given again after the reaction has improved, and the reaction is observed. Recurrence may support causality. Rechallenge is not routinely performed only to prove causality and must never be attempted without an appropriate clinical decision.

Confounding factors

Other factors that may explain the reaction, such as an underlying disease, another medicine, an infection, food interaction, or an environmental exposure.

4. Information Needed for Assessment

- Patient details relevant to the event, such as age and clinical history.
- Name of the suspected medicine, dose, route, frequency, start date, and stop date.
- Description of the reaction, onset date, severity, and outcome.

- Time relationship between medicine use and the reaction.
- Other medicines being taken at the same time.
- Relevant laboratory results, investigations, and medical diagnoses.
- What happened after the suspected medicine was stopped or the dose was changed.
- Whether the reaction is already known for that medicine.

5. General Steps in Causality Assessment

- **Step 1 – Identify the suspected medicine:** Record all medicines used before the reaction.
- **Step 2 – Describe the reaction:** Document the symptoms, diagnosis, onset, severity, and outcome.
- **Step 3 – Check the time relationship:** Decide whether the reaction began after the medicine was started and whether the timing is biologically reasonable.
- **Step 4 – Review other possible causes:** Consider diseases, other medicines, and non-drug causes.
- **Step 5 – Assess dechallenge information:** Check whether the reaction improved after stopping or reducing the suspected medicine.
- **Step 6 – Review rechallenge information, if available:** Check whether the reaction returned after re-exposure. Rechallenge is often unavailable and should not be done solely for assessment.
- **Step 7 – Check previous knowledge:** Review whether the reaction is described in reliable medicine-safety information.
- **Step 8 – Assign a causality category:** Use a recognised method, document the reasoning, and report the ADR as required.

6. WHO–UMC Causality Assessment System

The World Health Organization–Uppsala Monitoring Centre (WHO–UMC) system is a commonly used method for classifying the likelihood that a medicine caused an adverse reaction.

Category	Meaning
Certain	A clear time relationship exists; the event is unlikely to be explained by disease or other medicines; the response to stopping the medicine is clinically reasonable; and the event is pharmacologically or medically convincing. Rechallenge may strengthen the evidence but is not always required.
Probable / Likely	A reasonable time relationship exists; other causes are unlikely; and improvement after stopping the medicine is clinically reasonable. Rechallenge is not required.
Possible	A reasonable time relationship exists, but the reaction could also be explained by disease or other medicines. Information about stopping the medicine may be unclear or unavailable.
Unlikely	The timing makes a relationship improbable, and another explanation is more likely.
Conditional / Unclassified	More information is essential for assessment, or the additional information is being examined.
Unassessable / Unclassifiable	The information is insufficient or contradictory, and it cannot be supplemented or verified.

7. Naranjo Algorithm

The Naranjo Algorithm is a structured questionnaire used to estimate the probability that an adverse event is related to a particular medicine. It contains 10 questions. Each answer is assigned a score, and the total score

determines the probability category.

Question (short form)	What is checked?
1	Are there previous conclusive reports of this reaction?
2	Did the reaction occur after the suspected medicine was given?
3	Did the reaction improve when the medicine was stopped or an antagonist was given?
4	Did the reaction reappear when the medicine was re-administered?
5	Are there alternative causes that could have caused the reaction?
6	Did the reaction reappear with placebo?
7	Was the medicine detected in toxic concentrations in body fluids?
8	Did the reaction become more severe with a higher dose or less severe with a lower dose?
9	Did the patient have a similar reaction to the same or a similar medicine previously?
10	Was the reaction confirmed by objective evidence?

Note: The Naranjo questions are scored as Yes, No, or Do not know. The score key differs by question, so use the standard Naranjo scoring table rather than assigning the same points to every "Yes."

Total score	Probability category
9 or more	Definite
5–8	Probable
1–4	Possible
0	Doubtful

8. Simple Example

A patient starts Medicine A. A few days later, the patient develops a skin rash. The clinician reviews the timeline, checks for infection and other medicines, and finds no more likely explanation. The rash improves after Medicine A is stopped.

Interpretation: The timing and improvement after withdrawal support a possible relationship. Depending on the complete clinical information, the case may be classified as "Probable / Likely" or "Possible" using WHO–UMC. A final category cannot be assigned from this short example alone.

9. Importance in Pharmacovigilance

- It supports consistent evaluation of suspected ADR reports.
- It helps distinguish a suspected association from a confirmed cause.
- It improves the quality of medicine-safety data.
- It supports signal detection and risk evaluation.
- It contributes to safer prescribing and patient counselling.

10. Quick Revision

- **Causality assessment:** Evaluation of the relationship between a medicine and an adverse reaction.
- **Dechallenge:** Observe the reaction after stopping or reducing the suspected medicine.
- **Rechallenge:** Observe the reaction after re-exposure; never do this solely to prove causality.

- **WHO–UMC:** Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified, Unassessable/Unclassifiable.
- **Naranjo:** A 10-question scoring algorithm; score categories are Definite, Probable, Possible, and Doubtful.

Suggested Study References

• WHO–UMC: The use of the WHO–UMC system for standardised case causality assessment.

• Naranjo CA et al. A method for estimating the probability of adverse drug reactions. Clinical Pharmacology & Therapeutics (1981).

• Standard pharmacovigilance and clinical pharmacy textbooks.

Study note: These are educational notes on the topic of causality assessment, not a verbatim transcript of the linked video. For a video-exact summary, share its transcript or captions. Clinical causality decisions should be made by qualified healthcare professionals using complete patient information.